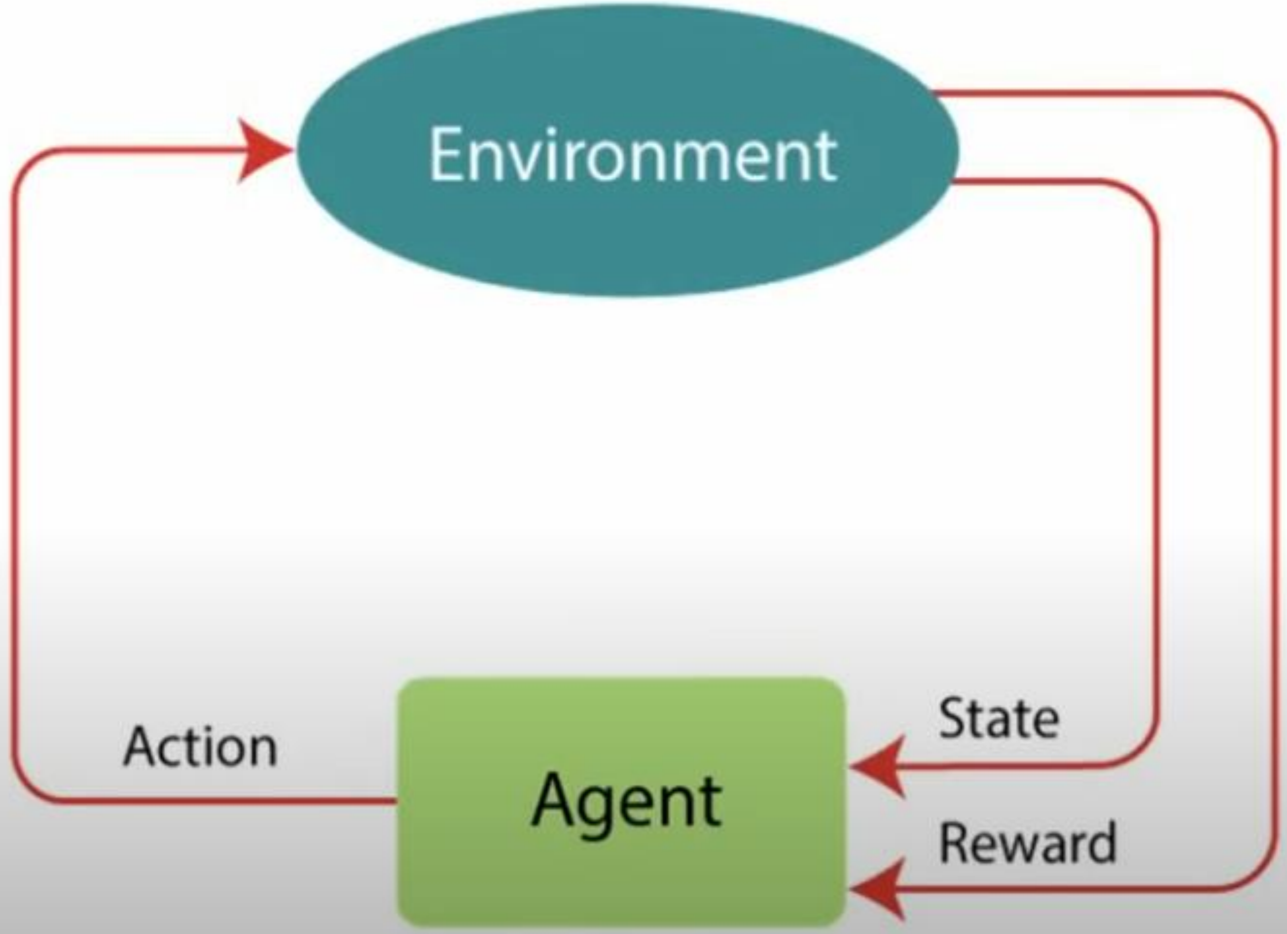
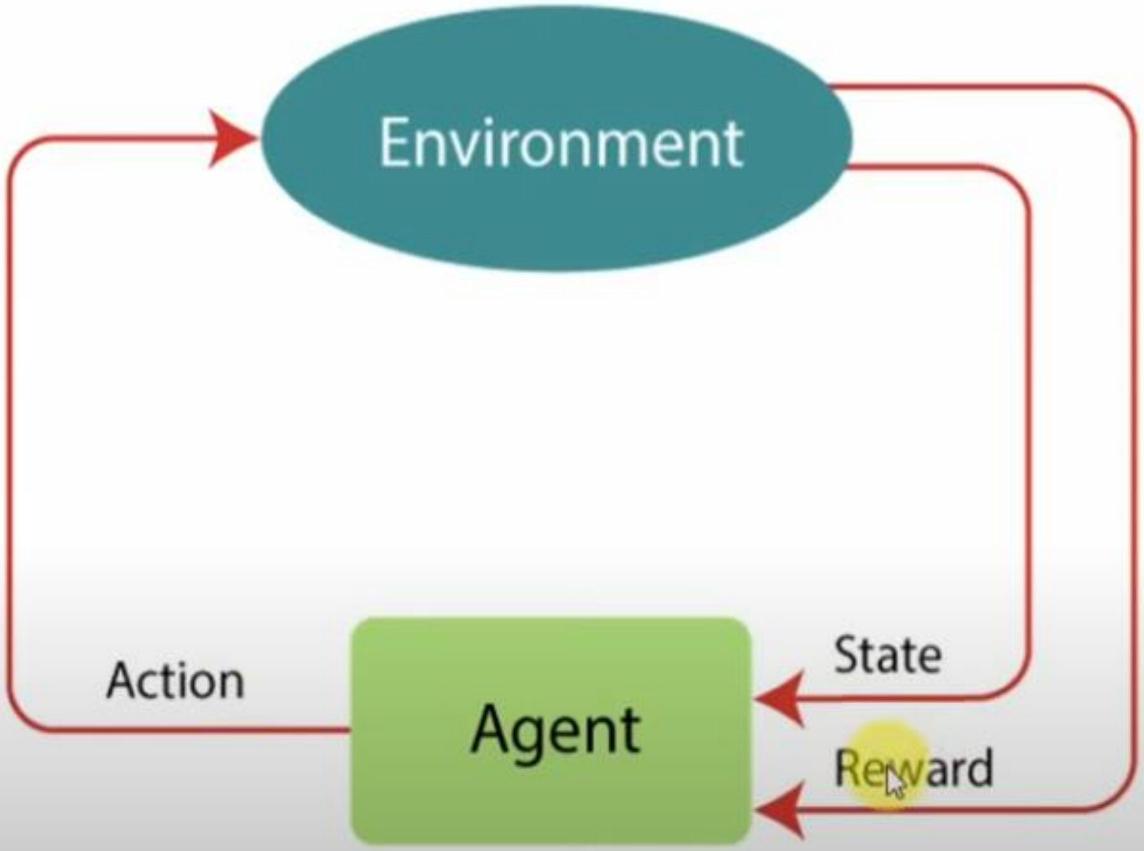
Reinforcement Learning



Introduction

Introduction to Reinforcement Learning

- Reinforcement Learning is a feedback-based Machine learning Approach here an agent learns to which actions to perform by looking at the environment and the results of actions.
- For each correct action, the agent gets positive feedback, and for each incorrect action, the agent gets negative feedback or penalty.



Introduction to Reinforcement Learning

- The agent interacts with the environment and identifies the possible actions he can perform.
- The primary goal of an agent in reinforcement learning is to perform actions by looking at the environment and get the maximum positive rewards.
- In Reinforcement Learning, the agent learns automatically using feedbacks without any labeled data, unlike supervised learning.
- Since there is no labeled data, so the agent is bound to learn by its experience only.
- Reinforcement Learning is used to solve specific type of problem where decision making is sequential, and the goal is long-term, such as game-playing, robotics, etc.

Introduction to Reinforcement Learning

- There are two types of reinforcement learning - **positive and negative.**
- **Positive reinforcement learning** is a recurrence of behaviour due to positive rewards.
- Rewards increase strength and the frequency of a specific behaviour.
- This encourages to execute similar actions that yield maximum reward.
- Similarly, in **negative reinforcement learning**, negative rewards are used as a deterrent to weaken the behaviour and to avoid it.
- Rewards decreases strength and the frequency of a specific behaviour.

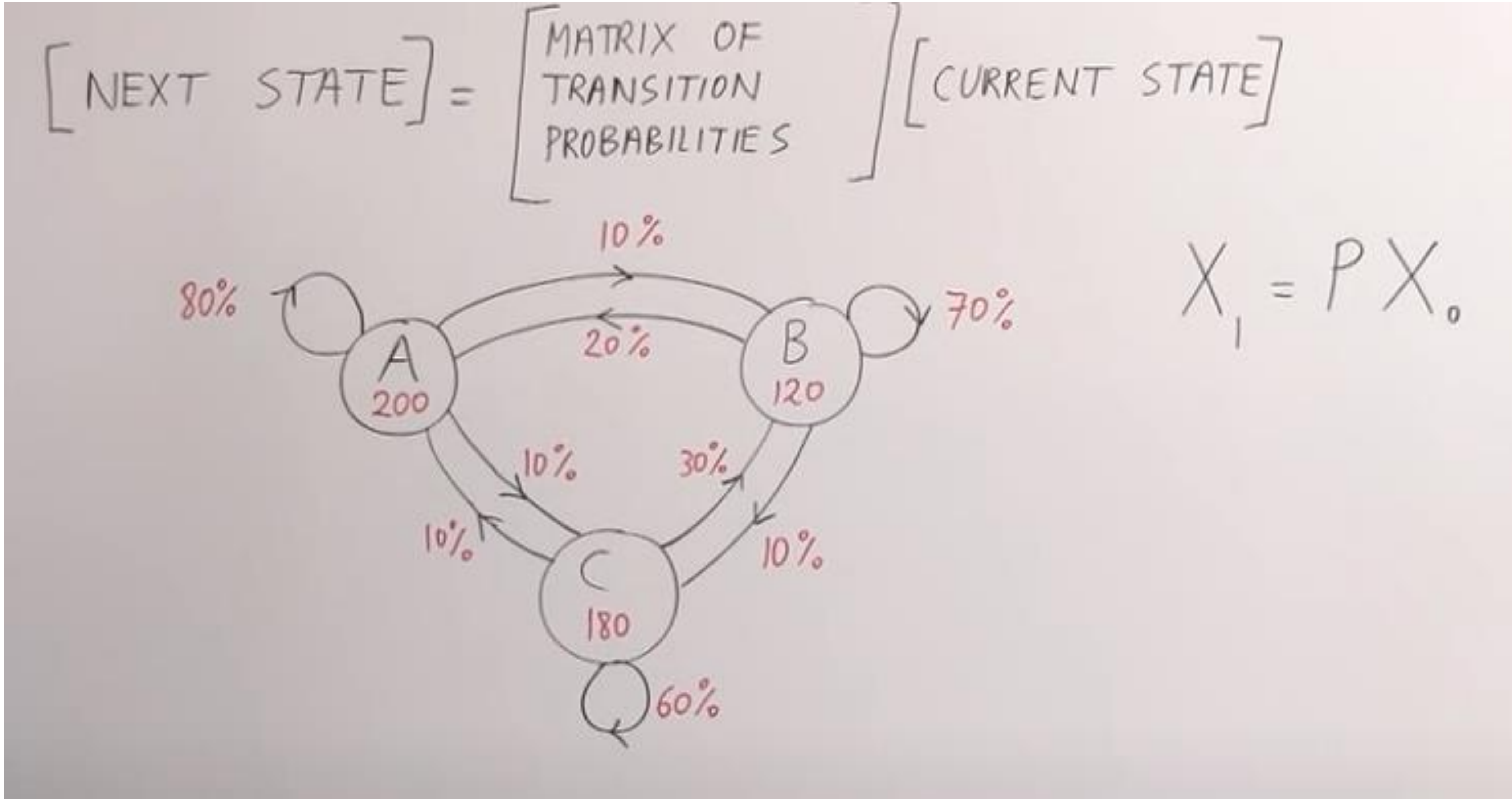
Introduction to Reinforcement Learning

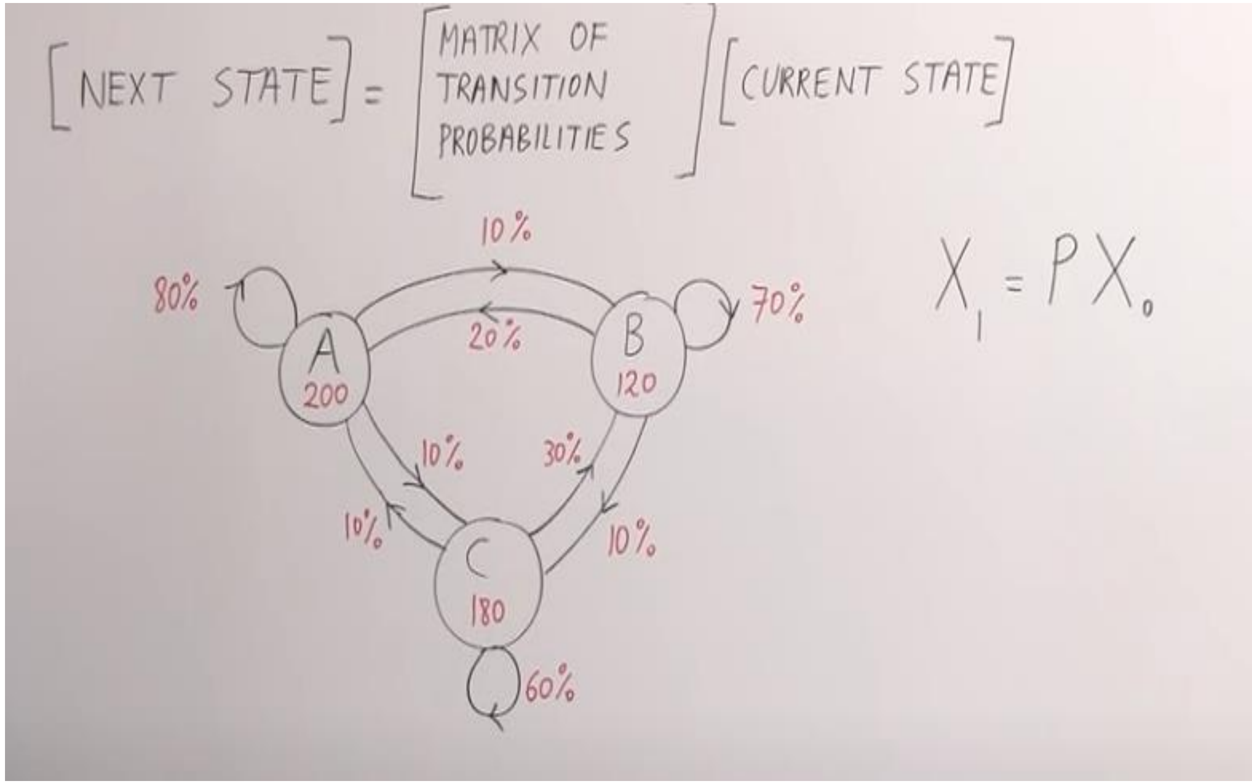
- In a **maze game**, there may a danger spot that may lead to loss.
- Negative rewards can be designed for such spots so that the agent does not visit that spot.
- Positive and negative rewards are simulated in reinforcement learning, say +10 for positive reward and -10 for some danger or negative reward.
- Reinforcement learning is an example of semi-supervised learning technique and is used to model sequential decision-making process.

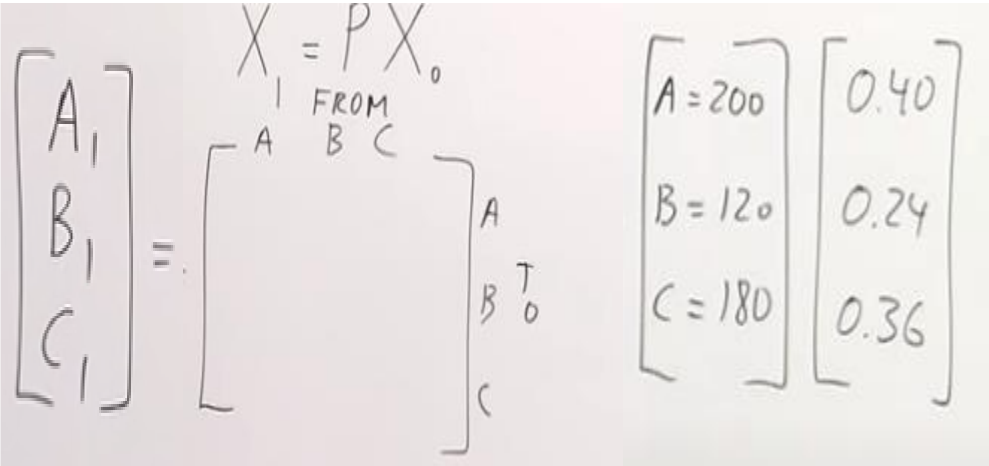
Scope of Reinforcement Learning

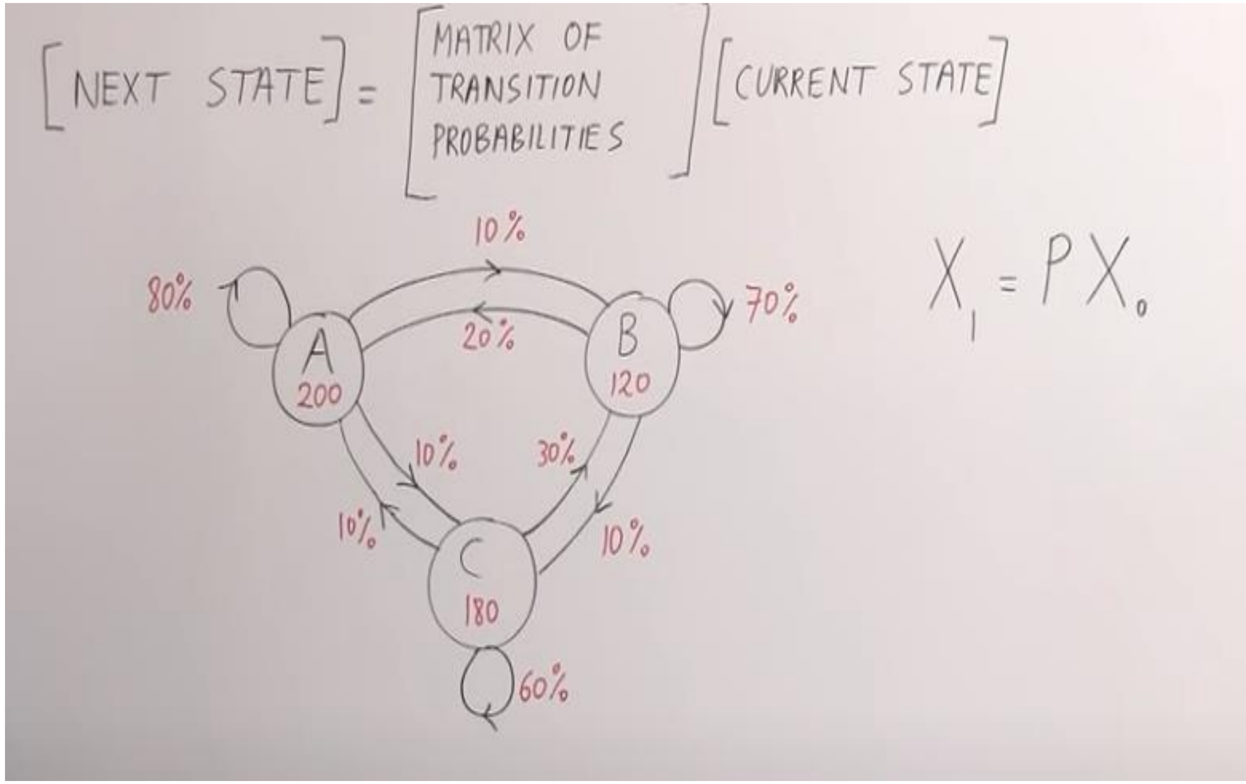
- Reinforcement is not suitable for environments where complete information is available.
- For example, the problems like object detection, face recognition, fraud detction can be better solved using a classifier than by reinforcement learning.

Markov Chains Example




[NEXT STATE] = [MATRIX OF TRANSITION PROBABILITIES] [CURRENT STATE]
10%
80%
70%
20%
A
200
B
120
10%
30%
10%
10%
C
180
60%
X1 = P X0


X1 = P X0
FROM
A B C
TO
A
B
C
A1
B1
C1
A = 200
B = 120
C = 180
0.40
0.24
0.36


[NEXT STATE] = [MATRIX OF TRANSITION PROBABILITIES] [CURRENT STATE]
10%
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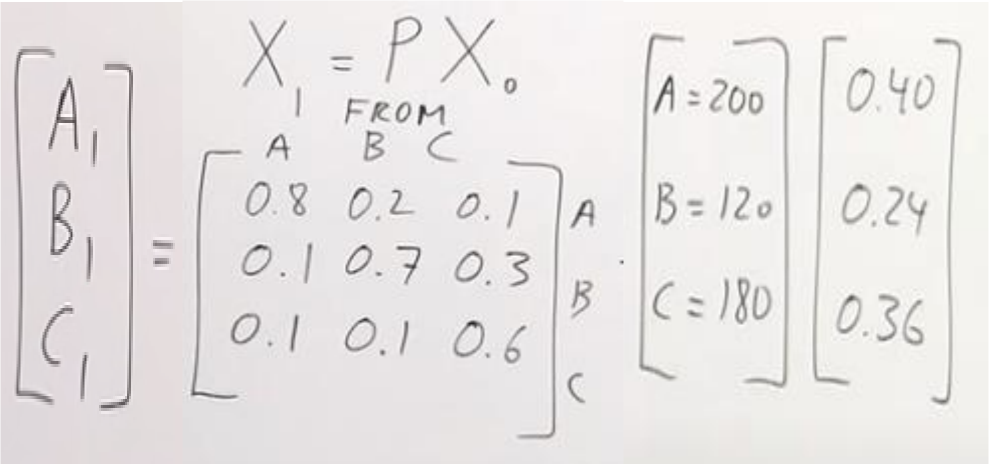
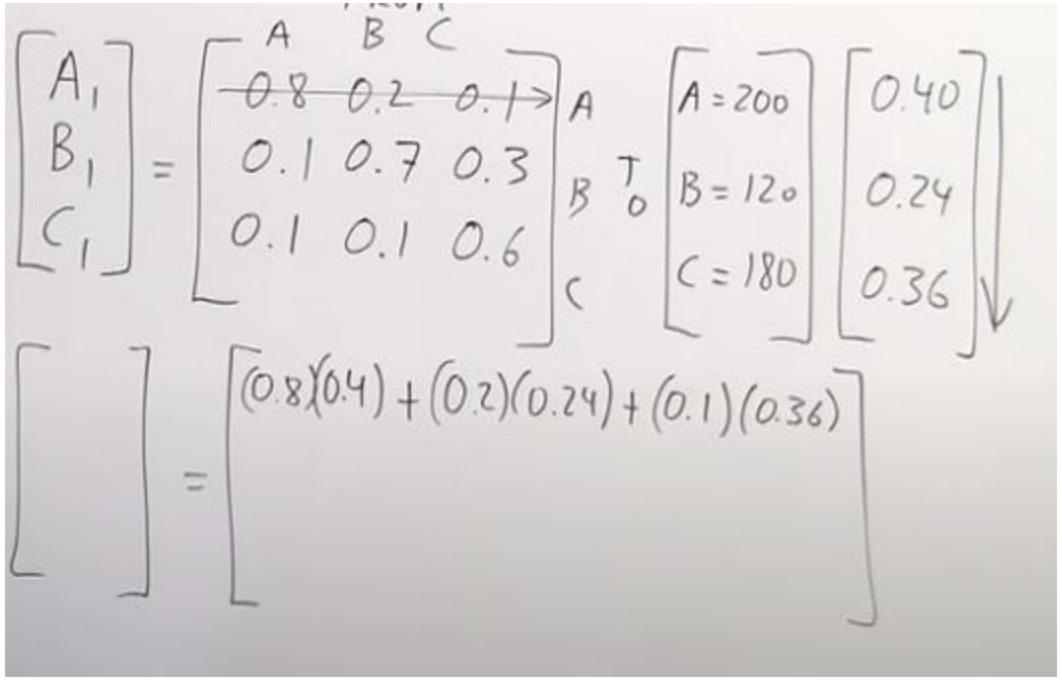
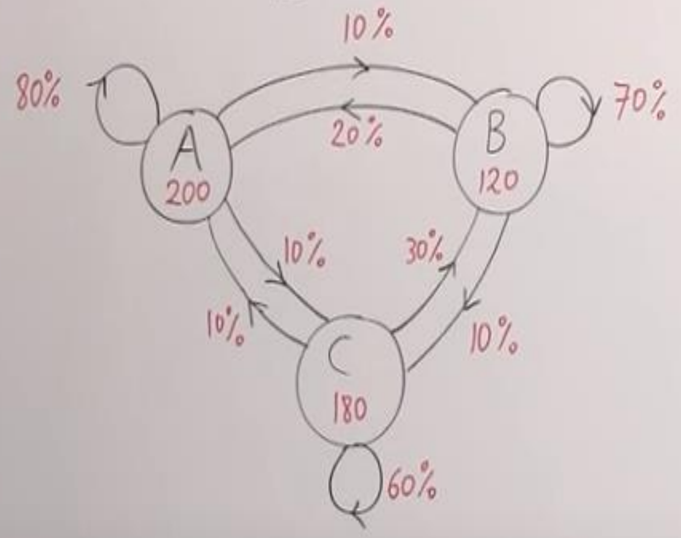
X1 = P X0


X1 = P X0
FROM
A B C
A1
B1
C1
=
0.8 0.2 0.1
0.1 0.7 0.3
0.1 0.1 0.6
A
B
C
A = 200
B = 120
C = 180
0.40
0.24
0.36


A B C
A1
B1
C1
=
0.8 0.2 0.1
0.1 0.7 0.3
0.1 0.1 0.6
A
B
C
T0
A = 200
B = 120
C = 180
0.40
0.24
0.36
=
(0.8)(0.4) + (0.2)(0.24) + (0.1)(0.36)

[NEXT STATE] = [MATRIX OF TRANSITION PROBABILITIES] [CURRENT STATE]



$$X_1 = PX_0$$

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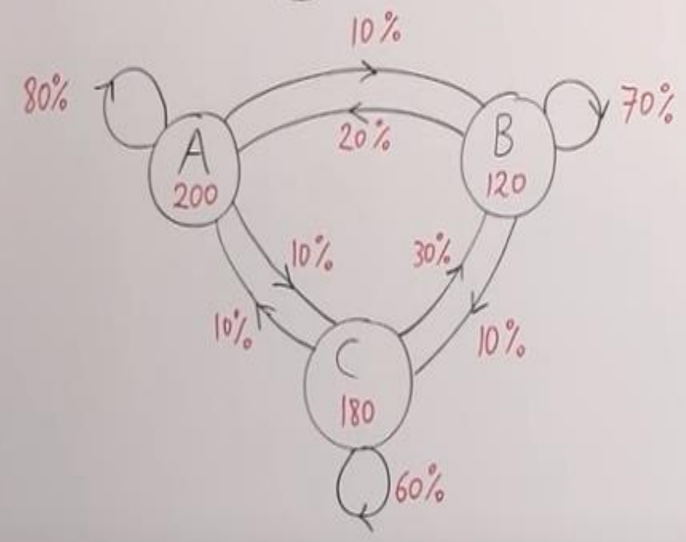
$$\begin{bmatrix} A_1 \\ B_1 \\ C_1 \end{bmatrix} = \begin{array}{c} \text{FROM} \\ \begin{array}{ccc} A & B & C \end{array} \\ \begin{bmatrix} 0.8 & 0.2 & 0.1 \\ 0.1 & 0.7 & 0.3 \\ 0.1 & 0.1 & 0.6 \end{bmatrix} \end{array} \begin{array}{c} A \\ B \\ C \end{array} \cdot \begin{bmatrix} A = 200 \\ B = 120 \\ C = 180 \end{bmatrix} \begin{bmatrix} 0.40 \\ 0.24 \\ 0.36 \end{bmatrix}$$

$$\begin{bmatrix} 0.404 \\ 0.316 \\ 0.280 \end{bmatrix} = \begin{bmatrix} (0.8)(0.4) + (0.2)(0.24) + (0.1)(0.36) \\ (0.1)(0.4) + (0.7)(0.24) + (0.3)(0.36) \\ (0.1)(0.4) + (0.1)(0.24) + (0.6)(0.36) \end{bmatrix}$$

1.000 ✓

$$[\text{NEXT STATE}] = \begin{bmatrix}\text{MATRIX OF} \\ \text{TRANSITION} \\ \text{PROBABILITIES}\end{bmatrix} [\text{CURRENT STATE}]$$



$$X_1 = PX_0$$

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$$\begin{bmatrix} A_1 \\ B_1 \\ C_1 \end{bmatrix} = \begin{array}{c} \text{FROM} \\ \begin{array}{ccc} A & B & C \end{array} \\ \begin{bmatrix} 0.8 & 0.2 & 0.1 \\ 0.1 & 0.7 & 0.3 \\ 0.1 & 0.1 & 0.6 \end{bmatrix} \end{array} \begin{array}{c} A \\ B \\ C \end{array} \; \text{TO} \; \begin{bmatrix} A = 200 \\ B = 120 \\ C = 180 \end{bmatrix} \begin{bmatrix} 0.40 \\ 0.24 \\ 0.36 \end{bmatrix}$$

$$X_1 = PX_0$$

$$\begin{bmatrix} A_1 \\ B_1 \\ C_1 \end{bmatrix} = \begin{array}{c} \begin{array}{ccc} A & B & C \end{array} \\ \begin{bmatrix} 0.8 & 0.2 & 0.1 \\ 0.1 & 0.7 & 0.3 \\ 0.1 & 0.1 & 0.6 \end{bmatrix} \end{array} \begin{array}{c} A \\ B \\ C \end{array} \; \text{TO} \; \begin{bmatrix} A = 200 \\ B = 120 \\ C = 180 \end{bmatrix} \begin{bmatrix} 0.40 \\ 0.24 \\ 0.36 \end{bmatrix}$$

$$\frac{\text{TOTAL}}{500} = \begin{array}{c} \# \\ \begin{bmatrix} 202 \\ 158 \\ 140 \end{bmatrix} \end{array} = \begin{bmatrix} 0.404 \\ 0.316 \\ 0.280 \end{bmatrix} = \begin{bmatrix} (0.8)(0.4) + (0.2)(0.24) + (0.1)(0.36) \\ (0.1)(0.4) + (0.7)(0.24) + (0.3)(0.36) \\ (0.1)(0.4) + (0.1)(0.24) + (0.6)(0.36) \end{bmatrix}$$

1.000 ✓

$$X_1 = PX_0 \qquad X_2 = PX_1 \qquad X_3 = PX_2 \quad \ldots\ldots$$